



Metabolic Symbiosis and Vulnerability in the CNS Axon-Myelin Unit

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Abstract

Axonal degeneration and demyelination are common pathological phenotypes in central nervous system (CNS) injury and degenerative diseases. Axonal and myelin diseases have generally been considered distinct phenomena, although our knowledge of why the axon-myelin unit (AMU), as a highly interconnected structure, serves as a common point of vulnerability to damage remains inadequate. In this review, we first reaffirm that the tight biological interconnections between axons and myelin should be recognized as a “symbiotic” unit across development, metabolism, and disease progression. Second, we systematically delineate the fundamental structural architecture and metabolic coupling of AMU. Third, we analyze how the impairment of one component of AMU influences the survival and functional integrity of others. In conclusion, we present a unique conceptual framework indicating that therapeutic methods targeting bioenergetic support in the CNS should concentrate on the energy coupling pathways and damage-crossover mechanisms of the AMU.

Keywords Axon-myelin unit · Metabolic coupling · MCT1 · Lactate shuttle · Axonal degeneration · Demyelination.

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Introduction

The human brain constitutes roughly 2% of the body’s mass while utilizing approximately 20% of total resting energy expenditure. (Engl and Attwell 2015). As the core unit of central electrochemical signal transmission, neurons are the dominant energy consumers, yet they lack significant energy storage capabilities. To cope with intense metabolic fluctuations in the central nervous system (CNS), neurons seem to rely on real-time energy supply and metabolic interconnection from surrounding glial cells. In the 1990s, (Pellerin et al. 1998) proposed the astrocyte–neuron lactate shuttle (ANLS) hypothesis, suggesting that glutamatergic synaptic activity triggers astrocyte glucose uptake and glycolysis, and that the generated lactate is shuttled to neurons via monocarboxylate transporters (MCTs) to provide metabolic support. The ANLS hypothesis first positioned astrocytes as the “energy stations” of neurons, opening up a new perspective for studying the energy-metabolic interaction between neurons and glial cells.

White matter tracts account for roughly half of the human brain and provide the anatomical substrate for “long-distance connections” in the vertebrate CNS. Remarkably, the adult human brain contains about 100 km of axons with extensive ramification and highly complex morphologies

(Steiner 2019). These axons utilize around 55% of the entire neuronal ATP reservoir to preserve resting potentials, propagate action potentials, and support synaptic vesicle cycling. (Zhang et al. 2022). Considering that axons in white matter are far from neuronal soma, and astrocyte contact with axons is generally spatially limited to some nodes of Ranvier, this raises an urgent question of how axons continuously obtain sufficient energy to maintain their metabolic normality and structural integrity.

Myelin sheaths encompass up to 98% of the CNS axonal surface and myelin-forming cell likely acts as a local fueling station for the axonal partner in white matter. In the peripheral nervous system (PNS), myelin is formed by Schwann cells. It forms continuous spirals of stair stepped or overlapping cytoplasmic expansions within compact myelin, namely Schmidt–Lanterman incisures (SLI). These incisures provide the structural basis for short-distance radial pathways that potentially facilitating the delivery of energy metabolites to the periaxonal space (Segura-Anaya et al. 2015; Meier et al. 2004). SLI are common in the PNS, but rare in the CNS. In contrast, myelin is formed by oligodendrocyte (OL) with only sporadic longitudinal incisures in CNS (Baumann and Pham-Dinh 2001). Indeed, OL couple to astrocytes via gap junctions, forming a generalized glial syncytium (panglial syncytium) that ultimately delivers energy metabolites to axons through myelin (Giaume and Liu 2012). Therefore, the ANLS has been expanded to also include OL, giving rise to an “astrocyte–neuron–oligodendrocyte lactate shuttle” (ANOLS) ternary coupling energy paradigm.

Although the multicellular neuronal–glia energy network exists, the axon–myelin unit (AMU) should be regarded as the minimal functional entity for energy supply and information integration in white matter. Myelin is no longer viewed as an inert insulator but as a metabolically active energy reservoir. It conveys energy substrates, including lactate, glucose, and fatty acids, to the axons for prompt ATP synthesis, while also providing myelin-derived exosomes to create a prolonged paracrine nutritional pathway. (Nave et al. 2023). The disintegration of the AMU triggers a cascade of energy crisis and induces axonal degeneration in diverse CNS injuries and neurodegenerative disorders. Here, we therefore define the AMU as a structural and functional complex composed of the axonal compartment of the neuron and its surrounding OL-generated myelin sheath, which is an inseparable unit throughout development, metabolic homeostasis and disease. Next, we summarize the structural characteristics and energy–metabolic coupling mechanisms of AMU in the CNS. Furthermore, we investigate the interrelationship between axons and myelin from a fundamental cell biology standpoint.

AMU Structural Characteristics

The formation of AMU begins with highly selective recognition between the processes of mature OL and target axons. A single OL in the CNS can extend multiple processes to different axons simultaneously. Each process eventually meanders into a flat lamina, which then begins to wrap around the axon in an Archimedean spiral, continuously enveloping it to form concentrically arranged myelin sheaths. Eventually, multiple layers of dense myelin lamellae are formed, a process known as the “liquid croissant” model (Sobottka et al. 2011). This process can be summarized into four consecutive stages: process recognition, lamina extension, spiral wrapping, and compression maturation (Kirby et al. 2006; Simons and Trotter 2007). In the CNS, each OL will form myelin sheaths around 20 to 60 axons (Chong et al. 2012). Morphologically, the myelin internode is composed of compact myelin and non-compact myelin.

When examined transversely, the cross-section of compact myelin reveals a distinctive periodic structure consisting of alternating electron-dense and electron-light layers. (Fig. 1A). The electron-dense layers, known as major dense lines (MDL), comprise the fused cytoplasmic surfaces of oligodendrocytes, which are abundant in myelin basic protein (MBP), facilitating the tight adhesion of myelin membrane leaflets. Conversely, the electron-dense layers (intraperiod lines, IPL) denote the closely aligned outer membranes, which are abundant in proteolipid protein (PLP) that sustains the exact IPL spacing. (Stadelmann et al. 2019) (Fig. 1B). The non-compact myelin comprises the paranodal loops, outer tongue, inner tongue, and sporadic longitudinal incisures within the compact myelin membranes. This intricate design seems somewhat more straightforward when myelin is illustrated in its unwrapped form as a vast network of interconnected intracellular compartments. (Edgar et al. 2021). The “myelin channels” provide a functional connection between the OL cell body and the inner myelin layer, allowing the rapid and direct transfer of energy-rich metabolites of OL to the myelinated axons (Fig. 1C).

Examined longitudinally, myelinated fibers are divided into the following structural domains: the node of Ranvier, the paranodes, the juxtaparanodes, and the internodes (Fig. 2). The node of Ranvier (approximately 0.8–1.1 μm in length) is a bare axolemma that separates adjacent myelin segments and contains voltage-dependent Na^+ channels at a high density. This organization facilitates swift ion exchange with the extracellular environment and energy-efficient saltatory conduction of action potentials. (Aggarwal et al. 2011). The adjacent paranodes (approximately 4 μm in length) form septate cellular junctions between the axolemma and the terminal ends of the myelin layers,

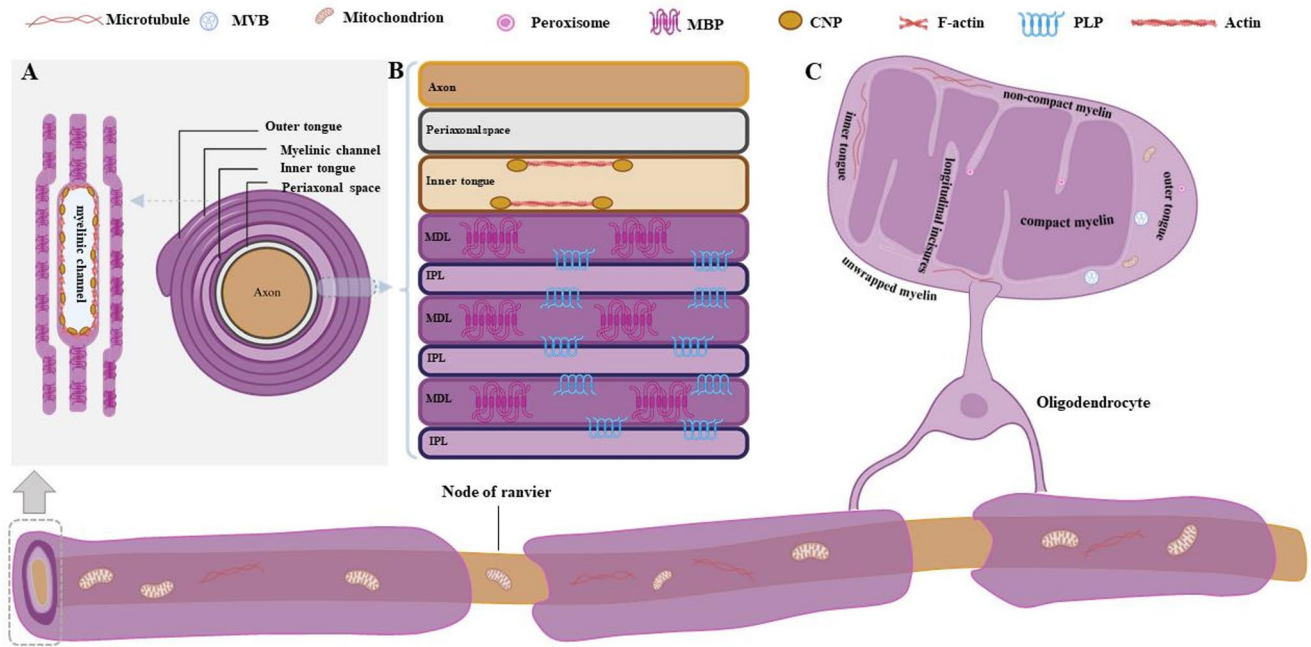


Fig. 1 Schematic diagram of the general structure of the axon-myelin unit. **A** Transverse section diagram of a myelinated axon illustrating the myelin channels (bright) and compacted myelin (dark). **B** The fundamental architecture of compressed myelin membranes, constituting the MDL and IPL, is facilitated by MBP and PLP, respectively. CNP within the inner tongue helps prevent myelinic channel collapse. **C** Schematic representation of a myelin-forming oligodendrocyte

in an unwrapped state, revealing non-compact myelin (light) and compacted myelin (dark) regions. The myelinic channel establishes a functional connection between the oligodendrocyte cell body and axons. MBP: myelin basic protein; PLP: proteolipid protein; CNP: 2',3'-cyclic nucleotide 3'-phosphodiesterase; MDL: major dense line; IPL: intraperiod line; MVEs: multivesicular endosomes

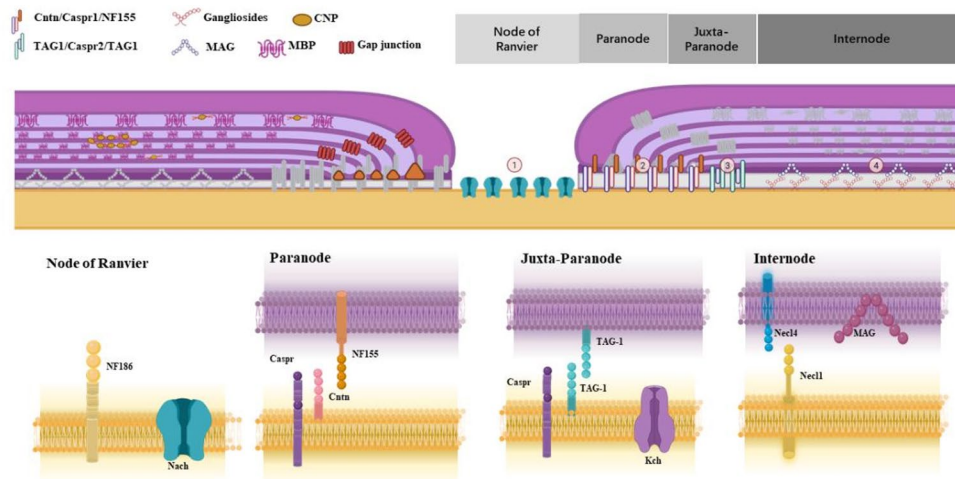


Fig. 2 Schematic diagram of the molecular composition of the axon-myelin unit. The myelinated axon shows the different domains, which are characterized by the expression of specific proteins. MBP helps generate compacted myelin membrane stacks, and CNP interacts with actin to form a myelinic channel. Gap junctions connect the paranodal loops of myelin. CNP: 2',3'-cyclic nucleotide 3'-phosphodiesterase;

MBP: myelin basic protein; MAG: myelin-associated glycoprotein; NF186: neurofascin-186; NaCh: voltage-gated Na^+ channels; KCh: voltage-gated K^+ channels; Caspr: contactin-associated protein; Cntn: contactin-1; NF155: neurofascin-155; Nectin1: Nectin-like molecule 1; Nectin4: Nectin-like molecule 4

held together by neurofascin 155 (NF155) on the glial surface complexing with Caspr1 and contactin1 (Cntn1) on the axonal side. The paranodal loops inhibit lateral diffusion of axonal proteins and promote the localisation of

nodal ion channels. In addition, paranodal peroxisomes accumulate in adaxonal myelin, which participates in lipid metabolism (Baes and Aubourg 2009), and the paranodal angle and gap junction form a radial pathway that allows

the free diffusion of ions and neurotrophic factors into the periaxonal space (Salzer et al. 2008). Therefore, the paranodes provide widespread ionic homeostatic regulation and energetic metabolism of the AMU. The juxtaparanodes (approximately 10–15 μm in length) are rich in voltage-gated K^+ channels and include a complex formed by the glial cell adhesion molecule TAG-1 and axonal adhesion proteins Caspr2. This assembly facilitates fast repolarisation and preservation of the domain architecture at the node of Ranvier. (Poliak et al. 2003; Saint-Martin et al. 2019; Traka et al. 2003). The internodes, measuring roughly 50 to 200 μm in length, possess many axonal membranes that do not directly interface with the extracellular environment but are oriented towards the insulating myelin sheath.

AMU Energy Coupling

Myelin in the CNS forms a compact insulating sheath that prevents ion leakage, effectively eliminating “electrical crosstalk” between adjacent axons. Due to its low capacitance and high resistance, myelin confines action potentials to the node of Ranvier between myelinated segments. This saltatory conduction significantly enhances the propagation velocity of action potentials along the axon. Additionally, myelination reduces the axonal capacitance and markedly lowers the energy cost of restoring resting ion gradients after each depolarization. Consequently, rapid conduction, fidelity in long-distance signal transmission, and spatial efficiency are the three primary advantages that myelin provides to the vertebrate nervous system. (Nave 2010; Stadelmann et al. 2019).

Although myelination markedly reduces the overall energy cost of electrical conduction, axons still face substantial local metabolic demands. A single action potential (AP) at a node of Ranvier requires at least 10^6 ATP molecules, corresponding to a metabolic cost of approximately 3.5 pmol ATP per cm^2 of membrane area (Aiello and Bachy-Rita 2000; Neishabouri and Faisal 2011). Notably, a second longitudinal conducting pathway is formed by the periaxonal and paranodal submyelin spaces. This periaxonal nanocircuit represents an additional energy-consuming process that accompanies saltatory conduction (Gorgoulis et al. 2019). Collectively, electrical signaling accounts for the second largest metabolic cost after synaptic transmission (Hallermann et al. 2012). Indeed, most of this ATP is consumed by Na^+/K^+ -ATPase to restore ionic gradients (Yang et al. 2023; Shrivastava et al. 2020). Specifically, Na^+/K^+ -ATPase actively expels Na^+ into the periaxonal space, with subsequent release at the paranode, establishing a return pathway from the axon to the endoneurial space that helps sustain ionic gradients (Trigo and Smith 2015). Loss

of axonal Na^+/K^+ -ATPase in the internode likely leads to continuous neurological decline in demyelinating diseases (Young et al. 2008). Facing high local energy requirements, axons must cooperate with the overlying myelin to secure essential metabolites. This metabolic dependence of axons on myelin has led to the idea of axon–myelin metabolic conjugation. We summarize many mechanisms of myelin trophic support for axons (Fig. 3).

Lactate Metabolism

Currently, the myelin-axon “lactate shuttle” model has received much attention, and myelin-derived lactate is crucial for long-term axonal integrity (Rinholm and Bergersen 2012). OL detects the metabolic requirements of axons through glutamatergic transmission and extracellular K^+ levels, subsequently regulating the rapid and immediate provision of lactate to axons to meet elevated energy demands. (Looser et al. 2024). OL absorbs and metabolises circulatory glucose to pyruvate by glycolysis, subsequently converting pyruvate to lactate. Such lactate in OL can also be directly taken up from the blood via the monocarboxylate transporter 1 (MCT1) (Fig. 4). Fünfschilling et al. (Fünfschilling et al. 2012) demonstrated the ability of OL to generate high levels of lactate for consumption by myelinated axons using conditional Cox10 mutant mice. When oxidative phosphorylation was explicitly inhibited in oligodendrocytes, these genetic models produced normal myelin. Moreover, magnetic resonance spectroscopy findings indicated elevated brain lactate concentrations in mutants compared to controls when isoflurane suppressed mitochondrial respiration. While brain lactate was rapidly metabolized within the white matter tracts and fell to undetectable levels upon washout of isoflurane (Fünfschilling et al. 2012). These results indicate that adult OL could maintain ATP levels with glycolysis alone and support that lactate flows from OL to neurons to maintain axonal integrity.

Notably, lactate in OL could also be obtained from “pan-glial syncytium” via gap junctions (Fig. 4A and B). Gap junctions are channels that link the cytoplasm of adjacent cells and allow rapid metabolic transport, electrical coupling, and spatial buffering. The “heterotypic gap junctions” have been observed between astrocytes and OL (Orthmann-Murphy et al. 2008). Such heterologous coupling would extend the size of a functional compartment from a single OL to a large glial network. Astrocytes take up glucose from the blood via glucose transporter 1 (GLUT1), store it as glycogen, and release lactate or pyruvate to the OL on demand. The OL/astrocyte gap junctions are composed of Cx32/Cx30, which mainly reside in the outer myelin sheaths, and Cx47/Cx43 that predominate at the OL soma (Nualart-Martí et al. 2013). These connexins are crucial for myelination and axonal

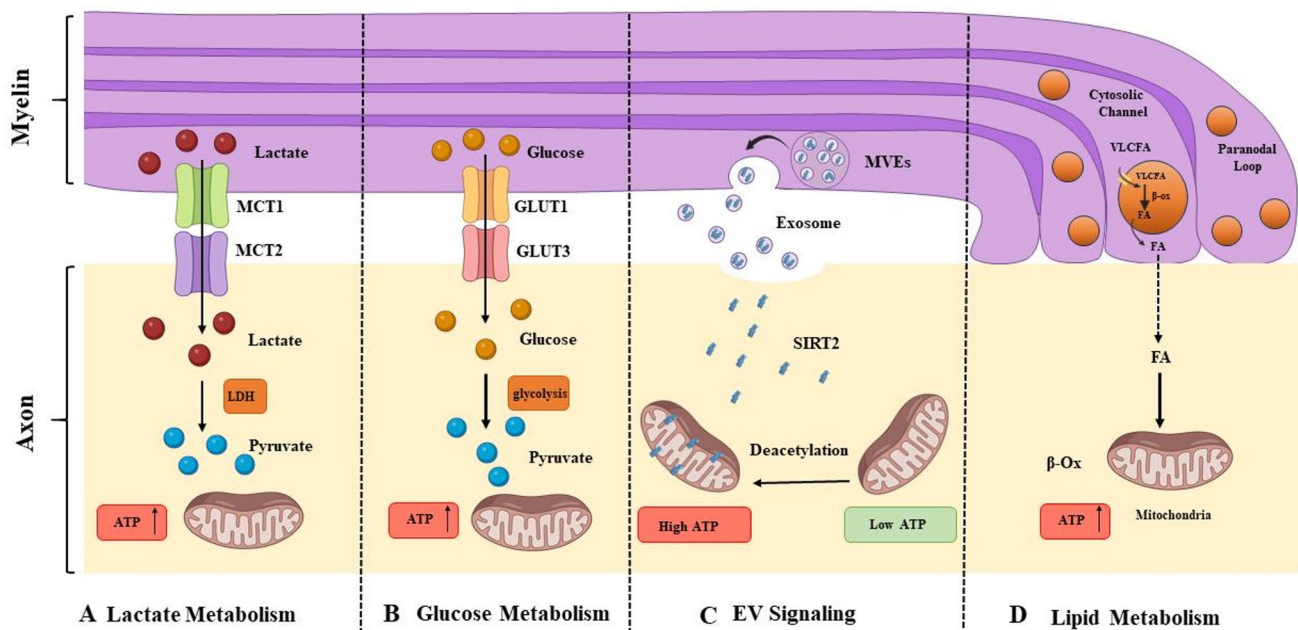


Fig. 3 The energy coupling mechanisms of the axon-myelin unit. **A** Lactate is transferred from myelin to axon via MCT1 and MCT2, where it is converted into pyruvate by LDH, resulting in ATP production. **B** In glucose metabolism, glucose is translocated via GLUT1 and GLUT3 from myelin to the axon, where it is transformed into pyruvate by glycolysis, resulting in ATP production. **C** In EV signalling, myelin releases MVE, and then the exosome-derived SIRT2 protein increases ATP production via deacetylation in axonal mitochondria. **D** In lipid metabolism, very long-chain fatty acids (VLCFA) undergo

peroxisomal β -oxidation in the paranodal loop, after which the resultant short-chain fatty acids are delivered to axonal mitochondria by a carnitine-dependent pathway for complete β -oxidation to generate ATP. MCT1: monocarboxylate transporter 1; MCT2: monocarboxylate transporter 2; LDH: lactic dehydrogenase; GLUT1: glucose transporter 1; GLUT3: glucose transporter 3; EV: extracellular vesicle; MVEs: multivesicular endosomes; SIRT2: sirtuin-2; VLCFA: very long chain fatty acids

viability, and mutations in these gap junctions result in demyelinating neuropathy in both the peripheral and central nervous systems. (Bruzzone et al. 1994; Kleopa et al. 2002; Orthmann-Murphy et al. 2007; Tress et al. 2011). Consequently, there is substantial justification to hypothesize a connection between lactate metabolism in oligodendrocytes and the activity of the “glial syncytium.”

Whether through intrinsic metabolism or glia coupling, the OL-derived lactate must rely on monocarboxylate transporters to feed axons (Figs. 3A and 4C). MCT1 at the adaxonal membrane of the myelin sheath exports lactate into the periaxonal space, whereas the monocarboxylate transporter 2 (MCT2) at the axolemma imports it to axons (Jha and Morrison 2020). In vitro studies have shown that suppression or downregulation of MCT1 expression in spinal cord organotypic slice cultures significantly reduces the survival rate of motor neurons and their axons, and this injury can be rescued by exogenous L-lactate supplementation under concurrent glucose deprivation (Lee et al. 2012). Thus, MCT1 knockdown blocks lactate export from OL without impeding its axonal uptake. In vivo studies have revealed that MCT1 heterozygous null mice eventually develop extensive axonal pathology in both the brain and spinal cord by 8 months of age (Lee et al. 2012). Since MCT1 is also expressed in

astrocytes, endothelial cells, and certain neuronal populations, it is crucial to further investigate the cell-type-specific conditional deletion of MCT1 using genetic approaches. Thomas Philips et al. (Philips et al. 2021) found that deletion of MCT1 in the OL lineage leads to late-onset axonal degeneration and hypomyelination in aged mice, whereas conditional knockout of MCT1 in astrocytes or endothelial cells does not result in neuronal damage (Suzuki et al. 2011). Collectively, these findings demonstrate that MCT1-mediated lactate metabolism is important for axonal energy homeostasis in the CNS.

Glucose Metabolism

Lactate is generally regarded as the principal energetic substrate for myelinated axons, because glucose is rapidly phosphorylated upon entry into OL (Baltan 2015). Myelin expresses GLUT1, while axonal membranes are abundant in GLUT3. Therefore, an alternative mechanism of trophic support for AMU has been proposed: OL may transfer glucose to the periaxonal area through Glut1, after which axons absorb glucose via Glut3. (Figures 3B and 4D). Trevisiol et al. [34] demonstrated that optic neurons perfused solely with lactate/pyruvate saw a significant decline in axonal

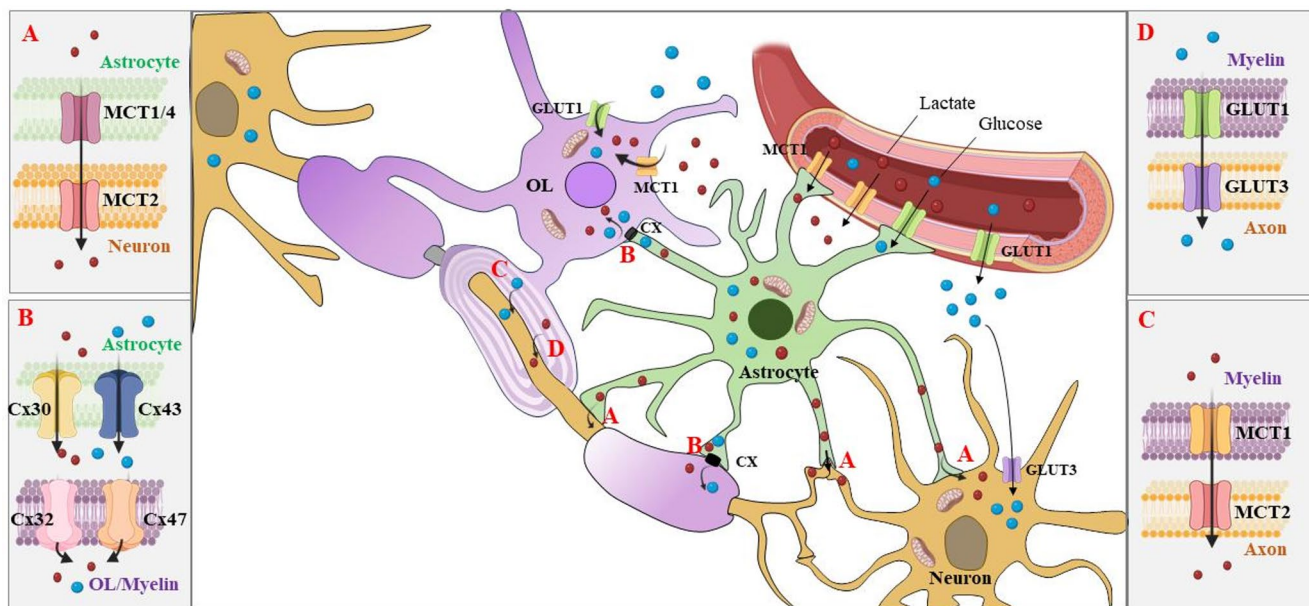


Fig. 4 Proposed metabolite shuttling between oligodendrocytes, astrocytes, and neurons. Glia cells (OL and astrocyte) import and metabolise blood glucose to pyruvate through glycolysis, then convert pyruvate to lactate. Such lactate in glia cells could also be directly taken up from the blood via MCT1. Astrocytes can transport lactate via MCT1/4 to neuronal soma, nodes of Ranvier, and synapses, whereas MCT2 imports it to neurons and converts it into ATP molecules to gen-

erate metabolic energy (A). Astrocytes can also transport lactate and glucose to OL via gap junctions that mainly reside in the outer myelin sheaths and OL soma (B). Then, lactate and glucose in OL are exported via transporters (MCT1, GLUT1) located in the myelin sheath and further imported into axons via transporters (MCT2, GLUT3), respectively (C and D)

ATP during high-frequency stimulation, suggesting that lactate/pyruvate alone is inferior to glucose in supporting ATP generation. Consequently, glucose may function as a crucial energy substrate to supply supplementary energy to axons.

The oligodendroglial supply of glucose to axons may differ across various parts of the central nervous system. In acute brain slices of the corpus callosum in mice, Niklas Meyer et al. (Meyer et al. 2018) observed that exogenous glucose deprivation (EGD) abolished compound action potentials (CAPs), and supplementation of lactate/pyruvate was unable to rescue them. In addition, loading OL with glucose effectively prevents CAP loss during EGD, whereas it failed in mice lacking CX47. Consequently, the authors suggested that callosal oligodendrocyte networks supply energy to maintain axonal activity, mostly via glucose transport. Future research is necessary to elucidate the specific processes of myelin–axon glucose coupling and to determine if glucose functions as a universal energy source for maintaining CNS axons.

Lipid Metabolism

When glucose is scarce during starvation, myelin may mobilize fatty acids to fuel axons (Fig. 3C). In *Drosophila*, cortical glia directly interface with neurons and use their lipid store to rescue energy-starved axons (Silva et al. 2022). In

mice, myelin fatty acid levels reduced under hypoglycemic conditions (Asadollahi et al. 2024). In humans, myelin was markedly reduced in specific brain regions after marathon running. (Ramos-Cabrera et al. 2025). Therefore, lipid use as a metabolic buffer during periods of extreme metabolic demand may be a conserved mechanism among different species. Although lipid droplets appear “lost” in axon-associated glial cells, lipid-rich myelin has evolved as a large ‘lipid store’ in vertebrates. Multi-layered myelin membranes comprise 70–75% lipid, and this substantial lipid content enables myelin to serve as a temporary energy storage. (Nave et al. 2017). The recent study supports this possibility. In states of starvation, oligodendroglial lipid metabolism serves as an energy store for axons, with the preservation of axonal function reliant on fatty acid β -oxidation occurring in peroxisomes within myelin. (Asadollahi et al. 2024). In addition, the genetic model with dysfunction of myelin peroxisomes in OL displayed myelin metabolic disturbance and serious axonal swelling in the CNS (Kassmann et al. 2011). Collectively, this research suggests that maintaining the myelin lipid metabolism is possibly the key to reversing axonal energy deficits in a metabolic crisis.

Currently, the detailed coupling mechanism of lipid metabolism of AMU has not been clarified. Peroxisomes are prevalent in adaxonal myelin and accumulate in the paranodal loops, positioning them optimally to facilitate

axonal metabolism. (Kassmann et al. 2011). When glucose availability declines, catabolism of myelin lipids releases free fatty acids (FA) that undergo peroxisomal β -oxidation to generate acetyl-CoA and short-chain FA. The short-chain FA can be delivered to axonal mitochondria through a carnitine-dependent pathway for full β -oxidation to maintain axonal energy supply. (Kassmann 2014). Meanwhile, acetyl-CoA could be utilized to produce ketone bodies within the cholesterol biosynthetic pathway. (Auestad et al. 1991; Geelen et al. 1983). Then, ketone bodies can be exported to axons through MCT1 and further undergo oxidative phosphorylation (OXPHOS) in axonal mitochondria to produce ATP (Asadollahi et al. 2024). Interestingly, mitochondria migrate through thin cytoplasmic channels and localize within the cytoplasm of paranodal loops (Nakamura et al. 2021). Focal netrin-1 signaling induces mitochondrial elongation and inner membrane hyperpolarization, and promotes glycolysis, thereby increasing lactate production to support membrane expansion and local axonal metabolism (Nakamura and Kennedy 2021).

Extracellular Vesicle (EV) Signaling

EV is in a novel mode of bidirectional axon-myelin energy linkage (Fig. 3D). Oligodendroglial multivesicular endosomes (MVEs) are the precursors of exosomes and accumulate at myelin channels and the periaxonal loops within the myelin sheath. Axonal electrical activity-dependent release of glutamate triggers oligodendroglial MVEs fusion and EV secretion through oligodendroglial AMPA and NMDA receptors. In turn, axons internalize the released exosomes by endocytosis (Frühbeis et al. 2013). In vitro study shows that OL-derived exosomes markedly increase neuronal viability under oxidative or metabolic stress (Fröhlich et al. 2014; Frühbeis et al. 2013). In vivo studies demonstrated that conditional Rab35 ablation in OL inhibited MVE fusion and reduced exosome release, resulting in increased susceptibility of axons to oxidative damage (Hsu et al. 2010; Mukherjee et al. 2020). Experiments conducted in the 1980s on squid gigantic axons yielded the initial evidence of vesicular exchange between adaxonal glia and the axon, indicating an evolutionarily conserved mechanism of metabolic coupling between glia and axons. (Buchheit and Tytell 1992).

The oligodendroglial exosomes provide axons with key factors that enhance axonal transport (Krämer-Albers and Werner 2023), stress resistance (Mukherjee et al. 2020) and energy homeostasis (Krämer-Albers et al. 2007). Next, we highlight the metabolic mechanism by which OL supports axons via delivery of the NAD⁺-dependent deacetylase SIRT2. SIRT2 is highly enriched in myelinating OL and secreted within exosomes to mediate deacetylation of axonal

mitochondrial proteins and then stimulate axonal mitochondrial ATP production (Krämer-Albers 2021). An in vitro study showed that SIRT2 knockdown or SIRT2 deletion in OL fail to enhance axonal ATP supply (Chamberlain et al. 2021). Importantly, deficiency of SIRT2 in mice causes late-onset axonopathy and locomotor disability associated with redox and energy imbalance (Fourcade et al. 2017). In vivo delivery of OL-derived SIRT2 to axon could rescue axonal mitochondrial deficiency in SIRT2 KO mice (Chamberlain et al. 2021). Recently, mitochondria transfer via extracellular vesicles (EVs) has been identified as a possible physiological phenomenon under metabolic stress (Mozafari et al. 2026). Whether these communicating oligodendroglia EVs carry mitochondria components to axons remain to be investigated. Collectively, these findings highlight the role of oligodendroglial exosome secretion in glia-mediated trophic support for axons.

AMU Pathology

The AMU's most significant characteristic is the substantial proximity of two plasma membrane surfaces within the CNS. Both the neuron and oligodendrocyte must sustain disproportionately massive peripheral compartments, with a plasma membrane that is 100 times heavier than their cell bodies, via a susceptible conduit. Because both membranes of the AMU must sustain rapid electrical activity at sites far removed from their respective somata, the AMU is likely an "energy vulnerability point" in the CNS. Axonal degeneration and demyelination are common pathological hallmarks in CNS injury and neurodegenerative diseases. However, these pathologies are usually the result of the combined effects of pathogenic factors (genetic, infectious, immune and traumatic) on AMU. Therefore, it is difficult to distinguish the primary and secondary injury targets. To elucidate AMU disease, myelin and the ensheathed axons should be regarded as an interconnected functional unit, necessitating an inquiry into how the impairment of one component influences the viability of the other in AMU from a cellular biological standpoint.

The Myelin to the Axons "Outside-In" Injury Model

Secondary axonal degeneration after demyelination is a multi-dimensional pathological cascade that combines electrophysiological dysfunction, ionic homeostasis disruption, energy metabolism failure, and toxic microenvironment attack (Fig. 5). First, demyelination impairs the paranodal contact between the myelin and axons and instigates sodium channels to spread along the axolemma. The upregulation of sodium channels subsequently causes notably slower

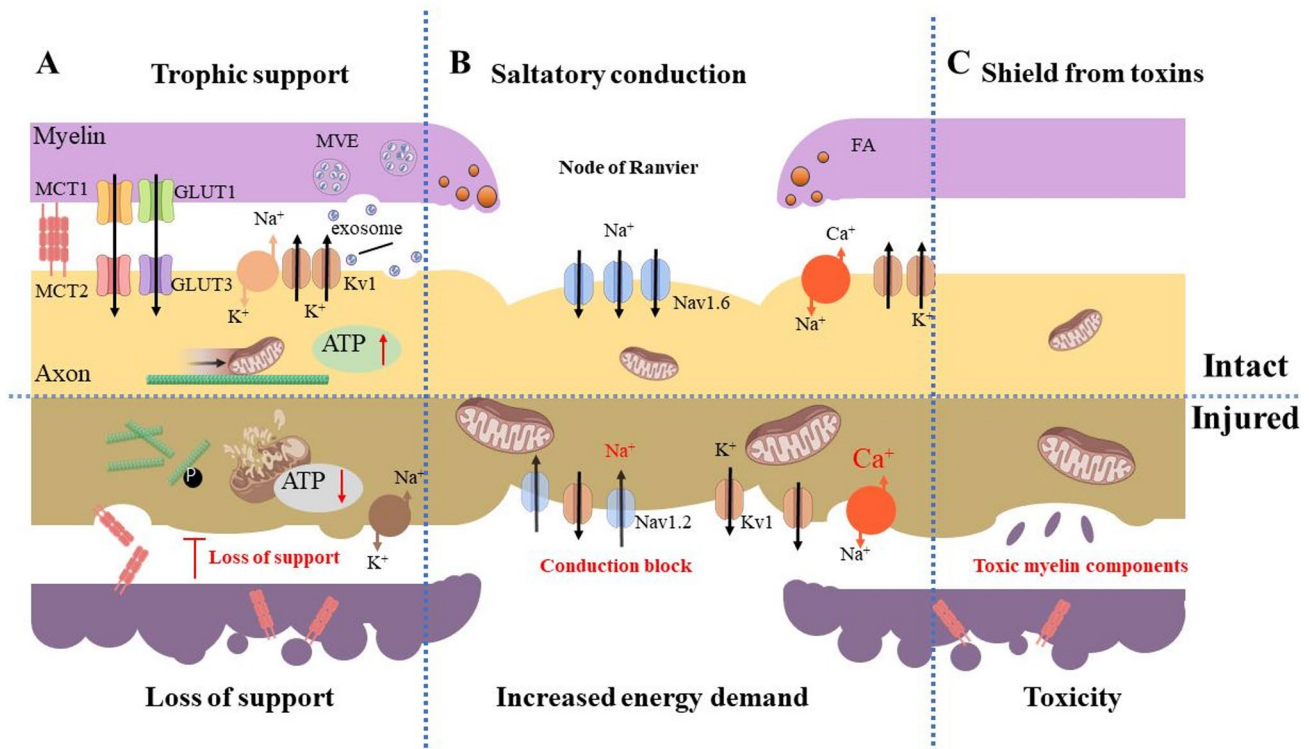


Fig. 5 Proposed mechanisms of coupling injury within the axon–myelin unit. **A** Loss of trophic or metabolic support from myelin. **B** Increased energy demand due to distributed sodium channels along the axolemma. **C** Gain of toxic myelin components

conduction and ectopic action potentials (Hamada and Kole 2015). Second, increased sodium channel expression results in increased operation of Na^+/K^+ -ATPase, the highly energetic process subsequently increases the number and volume of mitochondria. In addition, the $\text{Na}^+/\text{Ca}^{2+}$ -ATPase will start to operate in reverse mode to clear the increased axonal Na^+ load, which leads to accumulation of intra-axonal Ca^{2+} and activation of Ca^{2+} -dependent degradation pathways. Third, demyelination deprives the supply of trophic and metabolic energy substrates from myelin and OL, which further aggravates the energy burden. This condition has been designated as virtual hypoxia to signify a discrepancy between energy demand and supply. (Trapp and Stys 2009). Ultimately, demyelination probably results in the breakdown of the protective barrier afforded by myelin, exposing axons to deleterious myelin constituents.

Demyelinating animal models have provided pivotal insight into how myelin supports axons. Currently, four types of rodent models of demyelination paradigms have been developed: experimental autoimmune encephalitis (EAE) (Handel et al. 2011), toxin-induced models (Blakemore and Franklin 2008), viral infection models (Das Sarma 2010), and genetic models (Traka et al. 2010). We have summarized the principles of model induction, axon–myelin pathology, and the advantages and disadvantages of each model (Table 1). Toxin-induced models utilize the

application of chemicals with disruptive effects (cuprizone or lyssolecithin), but these toxic chemicals do not act to selectively injury myelin membranes, but directly disrupt axonal membranes (Pang et al. 2022; Praet et al. 2014). Experimental autoimmune encephalomyelitis and virus-induced demyelination depend on an immune response targeting a myelin antigen. Similarly, immune cells and the inflammatory environment can also directly attack axons (Constantinescu et al. 2011; Sato et al. 2011; Tsunoda and Fujinami 2002). Consequently, significant axonal degeneration occurs early in the first three models and even precedes the development of demyelination. In contrast, the gene knockout and genetic models eliminate the above potentially confounding non-cell-specific effects in a highly controlled and specific manner to induce demyelination without concurrent effects on axons. Consequently, these models may provide a more valuable framework for understanding the fundamental mechanisms governing axon–myelin interactions and clarifying the impact of demyelination on axonal integrity.

The myelin gene knockout model is generated through the germline knockout of genes exclusively expressed in the oligodendrocyte lineage, offering critical insights into the roles of proteins involved in myelin development and production (Brady et al. 1999; Edgar et al. 2009; Fruttiger et al. 1995; Griffiths et al. 1998; Kirkpatrick et al. 2001; Klugmann et al. 1997; Lappe-Siefke et al. 2003; Nguyen

Table 1 The pathogenesis, axon-myelin phenotype, advantages and disadvantages of demyelinating animal models

Models	Pathogenesis	Axon-myelin phenotype	Advantages of models	Disadvantages of models
EAE	Immunization with myelin glycoprotein peptides induces inflammatory demyelination	Impaired axon transport, increased axoplasmic calcium, irreversible cytoskeletal fragmentation, axon loss, and demyelination	Easy induction of the model and similar clinical course and autoimmune etiology with multiple sclerosis	Immune cells impair axonal transport and induce cytoskeletal insults directly, independently of demyelination
Cuprizone	Oral administration of the copper chelator cuprizone causes OL oxidative damage and demyelination	Axonal oedema, axonal degeneration, compromised axonal transport, enlarged mitochondrial stationary site size, oligodendrocyte death, and extensive demyelination in white matter	Ease of induction of wide demyelination	Sustained cytotoxicity and inflammatory state induced by cuprizone directly damage axons
Lyssolecithin	Focal injection of lyssolecithin functions as a detergent to damage membranes and provoke demyelination.	axonal swelling, axon loss, impaired axon transport, disrupted nodal architecture, focal demyelination	Directly produces obvious demyelination in restricted regions of injection	Lyssolecithin does not exclusively target myelin membranes; rather, it destroys axonal membranes, resulting in axonal injury.
TMEV	Viral infection damages OL and activates an immune response leading to demyelination	axonal swelling, decreased neurofilament phosphorylation, cytoskeletal fragmentation, widespread demyelination	Easy induction of the model and similar clinical course and autoimmune etiology with multiple sclerosis	Virus-mediated inflammatory environment directly causes axonal injury
Genic model	Genetic ablation of OL or myelin gene knockout	Various phenotypes of axon-myelin abnormalities	Prevents direct axonal injury caused by inflammation or toxins, advantageous for clarifying the mechanistic understanding of myelin's role in supporting axons.	Can only specifically identify the function of a specific protein in myelin

EAE: experimental autoimmune encephalitis; TMEV: Theiler's murine encephalomyelitis virus; OL: oligodendrocyte

et al. 2009; Pan et al. 2005; Rasband et al. 2005; Snaidero et al. 2017; Spörkel et al. 2002). Table 2 summarizes the axon-myelin pathology secondary to mutations in myelin-associated genes. Single myelin gene knockouts could lead to axon degeneration. For example, *Mbp* null mice lack compact myelin lamellae and exhibit reduced axonal caliber and diminished neurofilament phosphorylation (Brady et al. 1999; Kirkpatrick et al. 2001; Snaidero et al. 2017). Notably, some myelin gene knockout mice go on to develop severe defects in axonal transport and progressive axonal degeneration despite forming compact myelin (Edgar et al. 2009; Nguyen et al. 2009; Spörkel et al. 2002). The degenerative axonal phenotypes linked to myelin gene deletion indicate a crucial function of myelin in preserving axonal integrity. Progressive axonal loss may arise not only from evident demyelination but also from prolonged disruption of oligodendrocyte-axon connections.

Genetic models involve the ablation of mature oligodendrocytes in transgenic mice through the cell-specific action of diphtheria toxin (DT). The administration of DT induces primary oligodendrocyte cytotoxicity and considerable axonal damage (Ghosh et al. 2011; Oluich et al. 2012; Pohl et al. 2011). Strikingly, myelinated axons obviously survived longer than the targeted OL, indicating that the myelin compartment can feed axons independent of a viable glial soma. Myelin is a remarkably stable structure that may be near its thermodynamic equilibrium state. (Aggarwal et al. 2011). Upon ablation of oligodendrocytes, myelin does not undergo immediate disintegration; rather, it maintains structural integrity and continues to furnish metabolic support for axons through glycolytic enzymes and transporters. (Oluich et al. 2012; Pohl et al. 2011). After several weeks, the OL death case showed a subsequent degeneration of myelinated axons. Thus, the genetic models of OL ablation suggest that primary injury at the level of the cell body of OL could

Table 2 The axon-myelin pathology in myelin gene knockout models and genetic models of oligodendrocyte ablation

Animal models	Physiological functions	Myelin pathology	Axon pathology	References
MBP null	MBP: binds the cytoplasmic surfaces of the bilayers tightly together	Lack of compact myelin	Decreased neurofilament phosphorylation/axon caliber, reduced microtubule stability, minimal axon degeneration	(Brady et al. 1999; Kirkpatrick et al. 2001; Snaidero et al. 2017)
Plp1 null	PLP: adhesion of the extracellular leaflets of the myelin membrane	Compact myelin formation	Progressive axon loss, axonal spheroids containing multivesicular bodies, impaired axon transport, disorganized cytoskeletal components, axonal swelling	(Griffiths et al. 1998; Klugmann et al. 1997; Spörkel et al. 2002)
Mag null	MAG: communicates the signal to the axoplasm and induces the maturation of the axonal cytoskeleton	Compact myelin formation	Decreased axonal caliber, reduced neurofilament spacing and phosphorylation, and progressive axonal loss	(Frutiger et al. 1995; Nguyen et al. 2009; Pan et al. 2005)
Cnp null	CNP: implicated in the formation and maintenance of cytosolic channels	Compact myelin formation	Disrupted paranode architecture, progressive axon loss, axonal spheroids with accumulations of membranous organelles, axonal swelling, impaired axon transport, swollen inner tongues	(Edgar et al. 2009; Lappe-Siefke et al. 2003; Rasband et al. 2005)
OL ablation	OL forms myelin sheaths and supports axons	Late-onset myelin loss	Progressive impairment of axonal transport and axonal loss, disrupted nodal architecture	(Ghosh et al. 2011; Oluich et al. 2012; Pohl et al. 2011)

MBP: Myelin basic protein; PLP: Proteolipid protein; MAG: myelin-associated glycoprotein; CNP: 2',3'-cyclic nucleotide phosphodiesterase; OL: oligodendrocyte

spread to the connected myelin segments and further cause delayed axonal degeneration.

Notably, myelin is a deficient energy source in aging (Matute and Verkhratsky 2025), and studies in nonhuman

primates have shown reduced myelin levels and lipid alterations across various brain regions with aging (Xu et al. 2024; Dimovasili et al. 2025). Age-related myelin dysfunction typically involves myelin degenerative structural changes and alterations in myelin formation dynamics (Wu and Chen 2026). In aging and various brain disorders, myelin pathological morphology included reduced myelin thickness, formation of myelin balloons, and the accumulation of dark cytoplasm (Huang et al. 2025). Meanwhile, MCT1 expression is significantly reduced in aging, likely impairing monocarboxylate transport (Mayorga-Weber et al. 2025). Aging generates specific inflammatory transcriptome alterations and diminished proliferation and differentiation rates in oligodendrocyte progenitor cells (OPCs), resulting in the failure of OPCs to generate new myelin and maintain myelin integrity (Heo et al. 2024). Therefore, age-related myelin deterioration as a potential catalyst contributes to cognitive decline and neurodegeneration for various neurodegenerative conditions.

The Axons to the Myelin “In-Outside” Injury Model

Currently, the most thoroughly investigated form of axon loss is Wallerian degeneration (WD), which is defined as the progressive anterograde disintegration that occurs distal to a site of axonal or neuronal cell-body injury (Llobet Rosell and Neukomm 2019). Researchers now recognize Wallerian degeneration as a genetically regulated, autonomous axon-death program, rather than viewing it as a passive process caused by the cutting of the nutrient supply (Freeman 2014). WD could be at least divided into three morphologically discernible phases: acute axon degeneration (Phase I), lag phase in the distal axon (Phase II), and execution phase in the distal axon (Phase III) (Wang et al. 2012). In phase I, the axonal segments, both proximal and distal to the lesion, rapidly disintegrate for several hundred micrometres within 60 min. Such an early-onset degenerative event may be related to extracellular Ca^{2+} influx at the lesion site due to mechanical primary axotomy. In phase II, NAD^+ and ATP depletion impair energy homeostasis and then trigger a second, long-term Ca^{2+} wave. At this point, the distal axons exhibit a period of structural quiescence and maintain physiological functions. In phase III, the axon begins to disassemble, and microtubules start to break down, alongside the dismantling of the axonal skeleton. The axonal granular fragmentation is complete within 1–2 h from the onset.

The WD animal models in CNS generally are produced by spinal cord injury (Wang et al. 2009), transection of optic nerve (Alexandris et al. 2023), cortical injury (Mu et al. 2024), and retinal ganglion cellular damage (Saggu et al. 2010). Currently, a number of studies have proved that axonal degeneration could damage the structural integrity

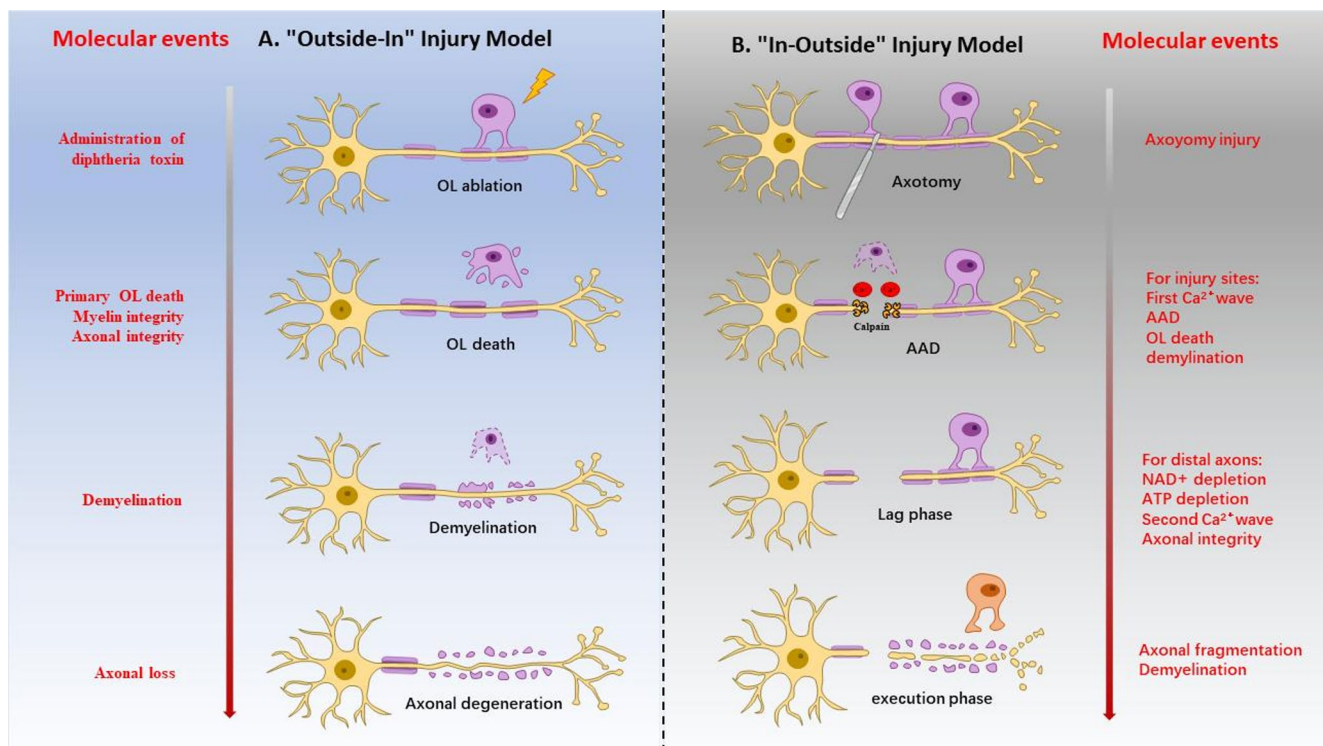


Fig. 6 Injury model of the axon–myelin unit. **A** “Outside-In” injury model. Primary injury targets the OL by administration of diphtheria toxin to cause OL ablation. In the early stage, myelin remains structurally stable and sustains support axons independent of a viable OL soma. In the late stage, myelin disintegrates and causes delayed axonal degeneration. **B** “In-Outside” injury model. After an axotomy injury, the local mechanical injury directly causes primary damage of all cellular components, resulting in OL death, AAD and demyelination in the axonal segments both proximal and distal to the lesion. Such an

early hyperacute degenerative event may be related to extracellular Ca^{2+} influx (first Ca^{2+} wave) at the lesion site. In the lag phase of WD, the distal axons maintain structural quiescence, whereas NAD^+ and ATP depletion impairs energy homeostasis and then triggers a second, long-term Ca^{2+} wave. In the execution phase of WD, axonal granular fragmentation occurs and subsequently causes myelin loss. OL: oligodendrocyte; AAD: acute axon degeneration; WD: Wallerian degeneration

of its myelin partner. In the early phase after axotomy, both axonal and myelin sheath disruptions are confined to the damage site as a result of the main mechanical trauma. (Alexandris et al. 2023). Several days post-injury, disease progresses to distal segments, where swollen axons exhibit pallor, axolemmal enlargement, and cytoskeletal disintegration, indicative of ‘watery degeneration’. In the chronic phase, it progressively exhibited ‘dark degeneration’, characterised by axoplasm containing an amorphous, granular, and dark substance. (Saggu et al. 2010). Simultaneously, their myelin partner showed various pathological changes, such as myelin vacuolation, myelin breakdown, thinner lamellae and formation of dense amorphous myelin profiles (Alexandris et al. 2023; Saggu et al. 2010; Wang et al. 2009). Therefore, axonal degeneration seems to transmit pathology to its myelin partner, yet the specific mechanism remains unclear.

WD triggers sustained microglial and astrocytic activation, yet its impact on the OL lineage remains highly controversial. Certain studies discovered significant oligodendrocyte death in the distal white matter subsequent to the

spinal cord damage. (Fan et al. 2023; Zhao et al. 2022). Oligodendrocyte apoptosis may occur due to the loss of axonal contact, which is crucial for oligodendrocyte survival during development. (Barres et al. 1993). In contrast, Li et al. (Li and Blakemore 2004) found that the number of myelin-supporting OL remained constant during WD. It seems that OL respond by entering an atrophy-like resting state when facing loss of survival signals and secondary inflammatory responses during WD in CNS (Vargas and Barres 2007). These markedly different conclusions might be related to the animal model, the distance between the sampling site and the lesion site, and the degree of injury. Indeed, only a limited number of animal research have sought to ascertain if WD impacts OL, and a definite conclusion is absent currently. In our opinion, cortical pyramidal neuron injury or retinal ganglion cellular damage maybe ideal animal models by investigation of WD in spinal CST or optical nerve. The initial injury affects neuronal somas while preserving their axons, so subtly mitigating the impact of direct mechanical stress on axon-myelin integrity and the disruption to the blood-brain barrier. Furthermore, future studies should also

investigate OL energy metabolism during the lag phase and function change during the execution phase, respectively.

Conclusions and Outlook

In summary, we delineate the structural and energetic coupling characteristics of the AMU. Axon and myelin disease under energy stress should be viewed as a highly interconnected single target rather than as separate entities, like AMU. The special geometry of axon-myelin interactions provides a framework for understanding patterns of pathology spread. If the original insult affects axons, the disease may extend to the myelin sheath and then to the oligodendrocyte soma, referred to as the “in-outside” injury paradigm. Conversely, if the main insult affects the oligodendrocyte soma, disease may propagate to the myelin sheath and subsequently to the ensheathed axons, referred to as the “outside-in” injury model. (Fig. 6). The precise mechanisms that govern axon degeneration and myelin loss remain unclear and require further exploration.

AMU injury is a hallmark of CNS injury and degenerative disease. Therapeutic therapies aimed at bioenergetic support in the central nervous system should also consider the axons and their metabolic interactions with myelin. Moreover, it is imperative to refine and progress methodologies for assessing energy levels in axons with high resolution, including in vivo single-cell-resolution imaging of the axon-myelin unit (AMU), high-resolution detection of ultrastructural details of the AMU, and real-time visualization of substrate flux across the axon-myelin interface. Further work is necessary to reveal the signals that mediate structural integrity and metabolic exchange in AMU. Further insights into the particular vulnerability of AMU enhance our comprehension of how a specific injury target might lead to distinctive pathological evolution of AMU in diverse CNS illnesses.

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Data Availability Data will be made available on request.

Declarations

Competing interests The authors declare no competing interests.

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